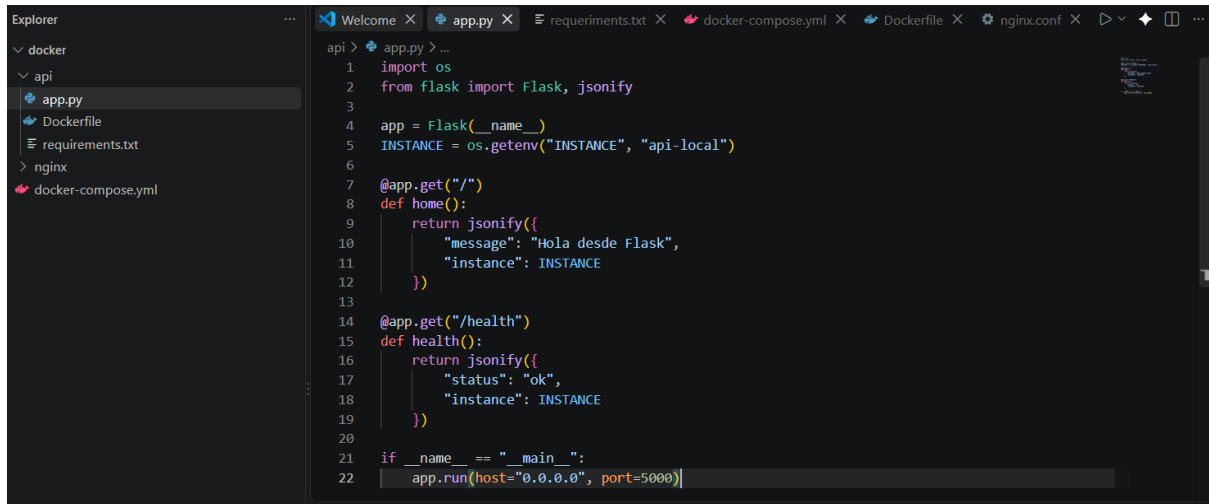
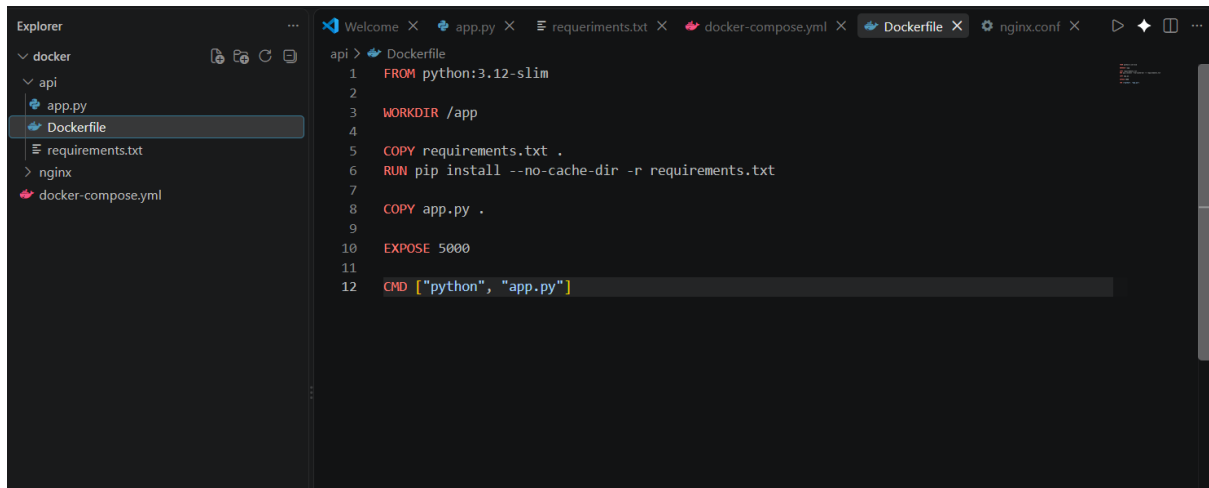


app.py



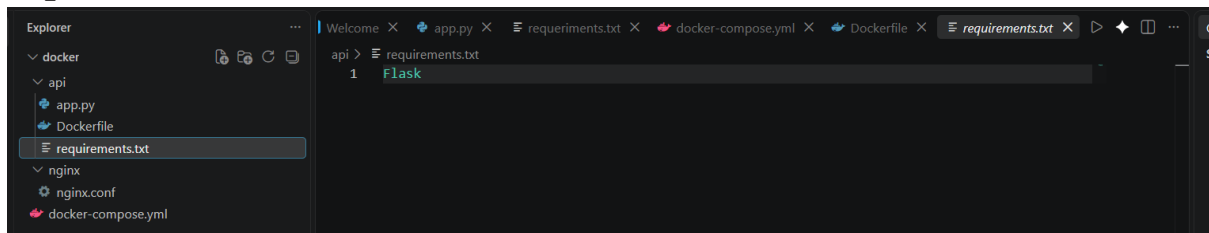
```
1 import os
2 from flask import Flask, jsonify
3
4 app = Flask(__name__)
5 INSTANCE = os.getenv("INSTANCE", "api-local")
6
7 @app.get("/")
8 def home():
9     return jsonify({
10         "message": "Hola desde Flask",
11         "instance": INSTANCE
12     })
13
14 @app.get("/health")
15 def health():
16     return jsonify({
17         "status": "ok",
18         "instance": INSTANCE
19     })
20
21 if __name__ == "__main__":
22     app.run(host="0.0.0.0", port=5000)
```

Dockerfile



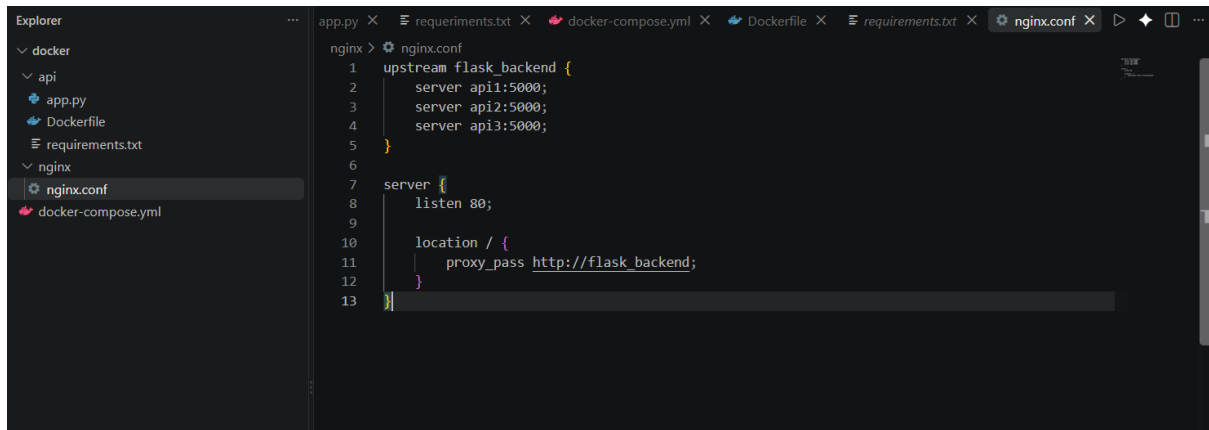
```
1 FROM python:3.12-slim
2
3 WORKDIR /app
4
5 COPY requirements.txt .
6 RUN pip install --no-cache-dir -r requirements.txt
7
8 COPY app.py .
9
10 EXPOSE 5000
11
12 CMD ["python", "app.py"]
```

requirements.txt



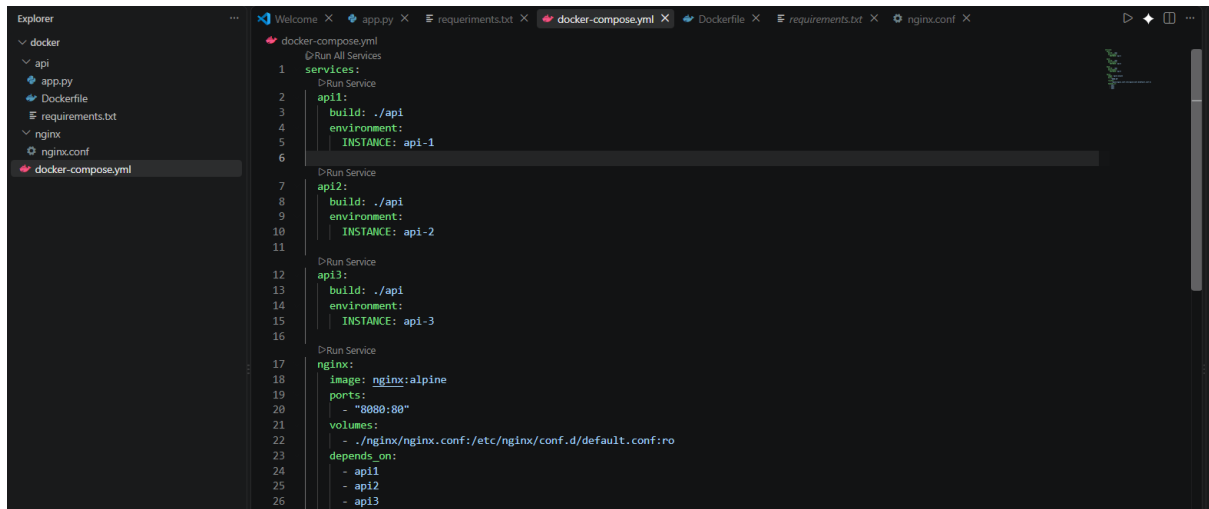
```
1 Flask
```

nginx.conf



```
1 upstream flask_backend {
2     server api1:5000;
3     server api2:5000;
4     server api3:5000;
5 }
6
7 server {
8     listen 80;
9
10    location / {
11        proxy_pass http://flask_backend;
12    }
13 }
```

docker-compose.yml



```
1 services:
2   api1:
3     build: ./api
4     environment:
5       INSTANCE: api-1
6
7   api2:
8     build: ./api
9     environment:
10      INSTANCE: api-2
11
12   api3:
13     build: ./api
14     environment:
15      INSTANCE: api-3
16
17   nginx:
18     image: nginx:alpine
19     ports:
20       - "8080:80"
21     volumes:
22       - ./nginx/nginx.conf:/etc/nginx/conf.d/default.conf:ro
23     depends_on:
24       - api1
25       - api2
26       - api3
```

Evidencia de compilación

docker compose up --build

```
Problems Output Debug Console Terminal Ports

00.9168" "-"
api1-1 | 172.19.0.5 - - [06/Sep/2026 03:23:02] "GET / HTTP/1.1" 200 -

api2-1 | 172.19.0.5 - - [06/Sep/2026 03:23:02] "GET / HTTP/1.1" 200 -

api1-1 | 172.19.0.5 - - [06/Sep/2026 03:23:02] "GET / HTTP/1.1" 200 -
nginx-1 | 172.19.0.1 - - [06/Sep/2026:03:23:02 +0000] "GET / HTTP/1.1" 200 50 "-" "Mozilla/5.0 (Windows NT; Windows NT 10.0; es-CO) WindowsPowerShell/5.1.261
00.9168" "-"

nginx-1 | 172.19.0.1 - - [06/Sep/2026:03:23:02 +0000] "GET / HTTP/1.1" 200 50 "-" "Mozilla/5.0 (Windows NT; Windows NT 10.0; es-CO) WindowsPowerShell/5.1.261
00.9168" "-"
nginx-1 | 172.19.0.1 - - [06/Sep/2026:03:23:02 +0000] "GET / HTTP/1.1" 200 50 "-" "Mozilla/5.0 (Windows NT; Windows NT 10.0; es-CO) WindowsPowerShell/5.1.261
00.9168" "-"
api3-1 | 172.19.0.5 - - [06/Sep/2026 03:23:02] "GET / HTTP/1.1" 200 -

nginx-1 | 172.19.0.1 - - [06/Sep/2026:03:23:02 +0000] "GET / HTTP/1.1" 200 50 "-" "Mozilla/5.0 (Windows NT; Windows NT 10.0; es-CO) WindowsPowerShell/5.1.261
00.9168" "-"
nginx-1 | 172.19.0.1 - - [06/Sep/2026:03:23:02 +0000] "GET / HTTP/1.1" 200 50 "-" "Mozilla/5.0 (Windows NT; Windows NT 10.0; es-CO) WindowsPowerShell/5.1.261
00.9168" "-"
nginx-1 | 172.19.0.1 - - [06/Sep/2026:03:23:02 +0000] "GET / HTTP/1.1" 200 50 "-" "Mozilla/5.0 (Windows NT; Windows NT 10.0; es-CO) WindowsPowerShell/5.1.261
00.9168" "-"
nginx-1 | 172.19.0.1 - - [06/Sep/2026:03:23:02 +0000] "GET / HTTP/1.1" 200 50 "-" "Mozilla/5.0 (Windows NT; Windows NT 10.0; es-CO) WindowsPowerShell/5.1.261
00.9168" "-"
nginx-1 | 172.19.0.1 - - [06/Sep/2026:03:23:02 +0000] "GET / HTTP/1.1" 200 50 "-" "Mozilla/5.0 (Windows NT; Windows NT 10.0; es-CO) WindowsPowerShell/5.1.261
00.9168" "-"
```

for (\$i=1; \$i -le 12; \$i++) { Invoke-RestMethod -Uri "http://localhost:8080/" }

```
PS C:\Users\esnei\Desktop\Ing. Sistemas\5 semestre\Computacion en la Nube\docker\taller1> for ($i=1; $i -le 12; $i++) { Invoke-RestMethod -Uri "http://localhost:8080/" }

instance message
-----
api-1   Hola desde Flask
api-2   Hola desde Flask
api-3   Hola desde Flask
api-1   Hola desde Flask
api-2   Hola desde Flask
api-3   Hola desde Flask
api-1   Hola desde Flask
api-2   Hola desde Flask
api-3   Hola desde Flask
api-1   Hola desde Flask
api-2   Hola desde Flask
api-3   Hola desde Flask
```

docker compose stop api2

```
PS C:\Users\esnei\Desktop\Ing. Sistemas\5 semestre\Computacion en la Nube\docker\taller1> docker compose stop api2
[+] stop 1/1
✓ Container taller1-api2-1 Stopped
PS C:\Users\esnei\Desktop\Ing. Sistemas\5 semestre\Computacion en la Nube\docker\taller1>
```

for (\$i=1; \$i -le 12; \$i++) { Invoke-RestMethod -Uri "http://localhost:8080/" }

```
PS C:\Users\esnei\Desktop\Ing. Sistemas\5 semestre\Computacion en la Nube\docker\taller1> for ($i=1; $i -le 12; $i++) { Invoke-RestMethod -Uri "http://localhost:8080/" }

instance message
-----
api-1   Hola desde Flask
api-3   Hola desde Flask
api-3   Hola desde Flask
api-1   Hola desde Flask
api-3   Hola desde Flask
api-1   Hola desde Flask
api-3   Hola desde Flask
api-1   Hola desde Flask
api-3   Hola desde Flask
api-1   Hola desde Flask
api-3   Hola desde Flask
api-1   Hola desde Flask
```

Invoke-RestMethod -Uri "http://localhost:8080/health"

```
PS C:\Users\esnei\Desktop\Ing. Sistemas\5 semestre\Computacion en la Nube\docker\taller1> Invoke-RestMethod -Uri "http://localhost:8080/health"

instance status
-----
api-1      ok
```